

Dataset: Online Learning Platform User Activity

```
activities = [  
    "LOGIN",  
    "COURSE_VIEW",  
    "LOGIN",  
    "VIDEO_PLAY",  
    "QUIZ_START",  
    "VIDEO_PLAY",  
    "LOGIN",  
    "COURSE_VIEW",  
    "ASSIGNMENT_SUBMIT",  
    "VIDEO_PLAY",  
    "QUIZ_START",  
    "QUIZ_COMPLETE",  
    "LOGIN",  
    "VIDEO_PLAY",  
    "COURSE_VIEW",  
    "LOGIN",  
    "VIDEO_PLAY",  
    "VIDEO_PLAY",  
    "QUIZ_START",  
    "LOGIN",  
    "COURSE_VIEW",  
    "COURSE_VIEW",  
    "ASSIGNMENT_SUBMIT",  
    "LOGIN",  
    "VIDEO_PLAY",  
    "QUIZ_COMPLETE",  
    "LOGIN",  
    "COURSE_VIEW",  
    "VIDEO_PLAY",  
    "VIDEO_PLAY",  
    "LOGIN",  
    "COURSE_VIEW",  
    "QUIZ_START",  
    "ASSIGNMENT_SUBMIT",  
    "LOGIN",
```

```
"VIDEO_PLAY",  
"LOGIN",  
"COURSE_VIEW",  
"QUIZ_COMPLETE",  
"VIDEO_PLAY",  
"LOGIN",  
"COURSE_VIEW",  
"VIDEO_PLAY",  
"LOGIN",  
"QUIZ_START",  
"COURSE_VIEW",  
"VIDEO_PLAY",  
"ASSIGNMENT_SUBMIT",  
"LOGIN",  
"VIDEO_PLAY"  
]
```

```
rdd = spark.sparkContext.parallelize(activities)
```

Basic Exercises

1. Display all activities.
2. Count total activity records.
3. Display first activity.
4. Display first 10 activities.
5. Display last activity using `collect()`.
6. Display unique activities.
7. Count unique activities.
8. Sort activities alphabetically.
9. Display activities in uppercase.
10. Display activity names and their lengths.

Filter Exercises

11. `Display only LOGIN activities.`
12. `Display only VIDEO_PLAY activities.`
13. `Display only QUIZ_START activities.`
14. `Display activities containing QUIZ.`
15. `Display activities ending with VIEW.`
16. `Remove LOGIN activities.`
17. `Remove VIDEO_PLAY activities.`
18. `Display activities longer than 8 characters.`
19. `Display activities shorter than 10 characters.`
20. `Display activities starting with C.`

Map Exercises

21. `Convert all activities to lowercase.`
22. `Add prefix EVENT_ to every activity.`
23. `Add suffix _2025 to every activity.`
24. `Convert activity names into their lengths.`
25. `Create tuples:`
`(activity, length)`
26. `Create tuples:`
`(activity, 1)`
27. `Replace LOGIN with USER_LOGIN.`

- 28. Replace VIDEO_PLAY with WATCH_VIDEO.
- 29. Reverse every activity string.
- 30. Extract first character from every activity.

Distinct Exercises

- 31. Find all distinct activities.
- 32. Count distinct activities.
- 33. Sort distinct activities.
- 34. Display distinct activities containing QUIZ.
- 35. Display distinct activities containing COURSE.

CountByValue Exercises

- 36. Count occurrences of each activity.
- 37. Find most frequent activity.
- 38. Find least frequent activity.
- 39. Display activity frequencies sorted by count.
- 40. Display activities occurring more than 5 times.

Union Exercises

```
extra_activities = [  
    "LOGIN",
```

```
"CERTIFICATE_DOWNLOAD",  
"VIDEO_PLAY",  
"PROFILE_UPDATE",  
"QUIZ_COMPLETE"  
]
```

```
rdd2 = spark.sparkContext.parallelize(extra_activities)
```

41. Union both RDDs.

42. Count records after union.

43. Display distinct records after union.

44. Count frequencies after union.

45. Display new activities introduced by second RDD.

Intersection Exercises

```
rdd3 = spark.sparkContext.parallelize([  
"LOGIN",  
"COURSE_VIEW",  
"PROFILE_UPDATE",  
"QUIZ_COMPLETE",  
"PAYMENT_SUCCESS"  
])
```

46. Find common activities between rdd and rdd3.

47. Count common activities.

48. Sort common activities.

49. Display common activities containing QUIZ.

50. Display common activities containing LOGIN.

Reduce Exercises

```
numbers = [  
120,  
250,  
300,  
450,  
600,  
750,  
900,  
1050,  
1200,  
1350,  
1500,  
1650,  
1800,  
1950,  
2100  
]
```

```
sales_rdd = spark.sparkContext.parallelize(numbers)
```

51. Find total using `reduce()`.
52. Find maximum using `reduce()`.
53. Find minimum using `reduce()`.
54. Find average using `reduce()` and `count()`.
55. Count numbers greater than 1000.
56. Double every value.
57. Add 100 to every value.
58. Sort ascending.
59. Sort descending.
60. Display top 5 values.

FlatMap Exercises

```
courses = [  
    "Python SQL Spark",  
    "Azure Databricks Spark",  
    "Python Pandas NumPy",  
    "SQL Spark DeltaLake",  
    "Azure Python SQL"  
]  
  
course_rdd = spark.sparkContext.parallelize(courses)
```

61. Split all words using flatMap().

62. Count total words.

63. Find distinct words.

64. Sort words alphabetically.

65. Count frequency of each word.

66. Display words containing Python.

67. Display words containing Spark.

68. Convert all words to uppercase.

69. Find longest word.

70. Find shortest word.

Real RDD Analytics

71. Find most common activity.

72. Find least common activity.

73. Find percentage of LOGIN activities.
74. Find percentage of VIDEO_PLAY activities.
75. Create activity report.
76. Create unique activity report.
77. Create frequency report.
78. Export distinct activities to a text file.
79. Export activity frequencies to a text file.
80. Generate final learning platform usage summary.